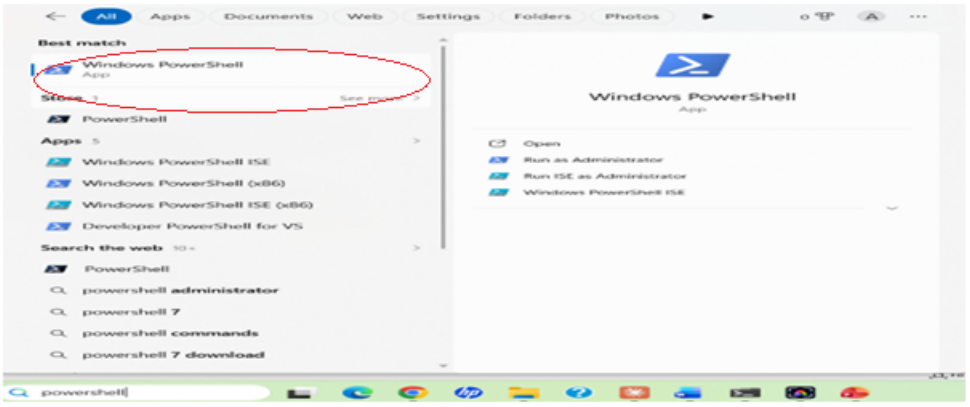
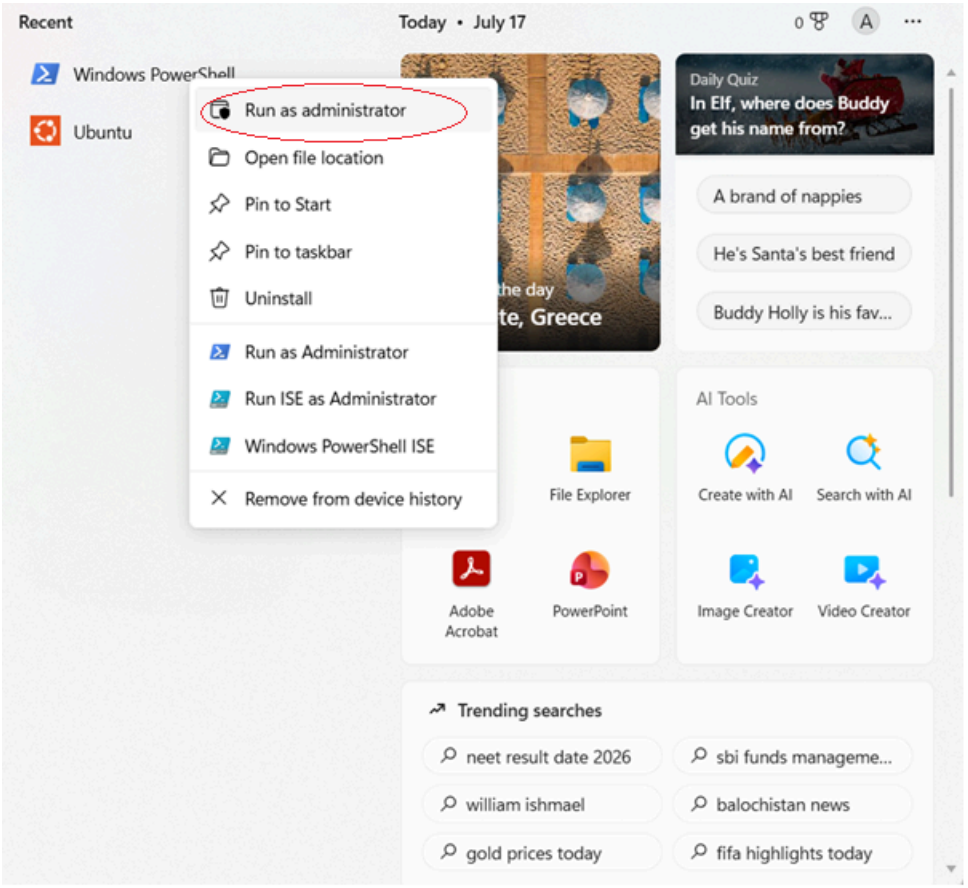
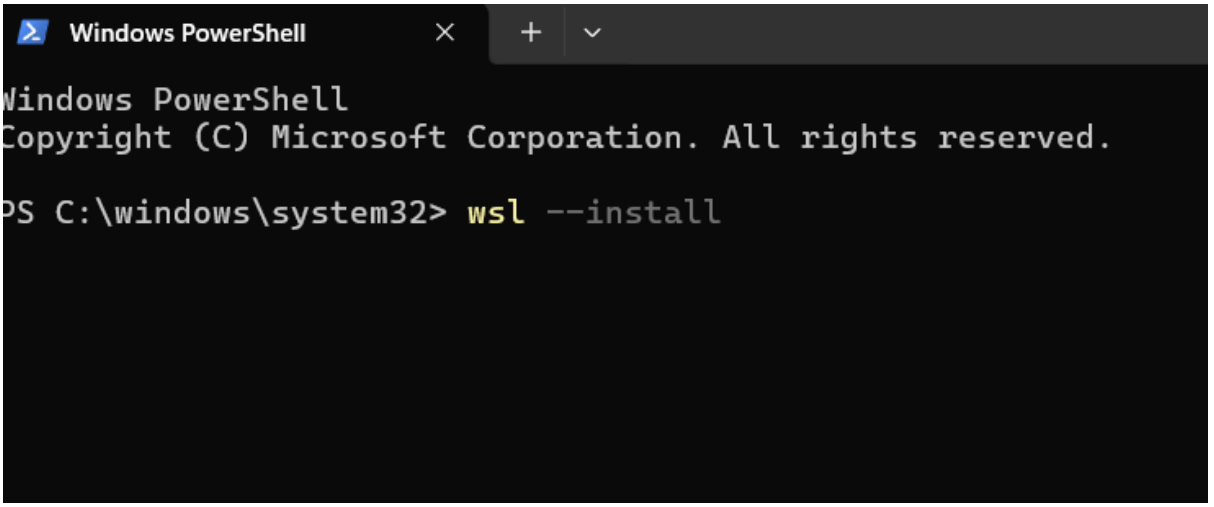
Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.





Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: anusha

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```
sanjana3027@DESKTOP-QJF5AGU:~$ sudo apt update
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Get:1 http://archive.ubuntu.com/ubuntu resolute InRelease [136 k
B]
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRele
ase [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main a
md64 Packages [417 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main T
ranslation-en [97.2 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main a
md64 Components [46.7 kB]
```

Step 7: Install GCC (C Compiler)

```
sanjana3027@DESKTOP-QJF5AGU:~$ sudo apt install build-essential
Upgrading:
  build-essential

Installing dependencies:
  libcrypt-dev

Summary:
  Upgrading: 1, Installing: 1, Removing: 0, Not Upgrading: 122
  Download size: 127 kB
  Space needed: 384 kB / 1024 GB available

Continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libcr
ypt-dev amd64 1:4.5.1-1 [122 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates/main amd
64 build-essential amd64 12.12ubuntu2.26.04.2 [5158 B]
Fetched 127 kB in 0s (305 kB/s)
Selecting previously unselected package libcrypt-dev:amd64.
(Reading database ... 42256 files and directories currently inst
alled.)
Preparing to unpack .../libcrypt-dev_1%3a4.5.1-1_amd64.deb ...
Unpacking libcrypt-dev:amd64 (1:4.5.1-1) ...
Preparing to unpack .../build-essential_12.12ubuntu2.26.04.2_amd
64.deb ...
Unpacking build-essential (12.12ubuntu2.26.04.2) over (12.12ubun
tu2) ...
Setting up libcrypt-dev:amd64 (1:4.5.1-1) ...
Setting up build-essential (12.12ubuntu2.26.04.2) ...
Processing triggers for man-db (2.13.1-1build1) ...
```

Step 8: Verify the Installation

```
sanjana3027@DESKTOP-QJF5AGU:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  T
here is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICUL
AR PURPOSE.
```

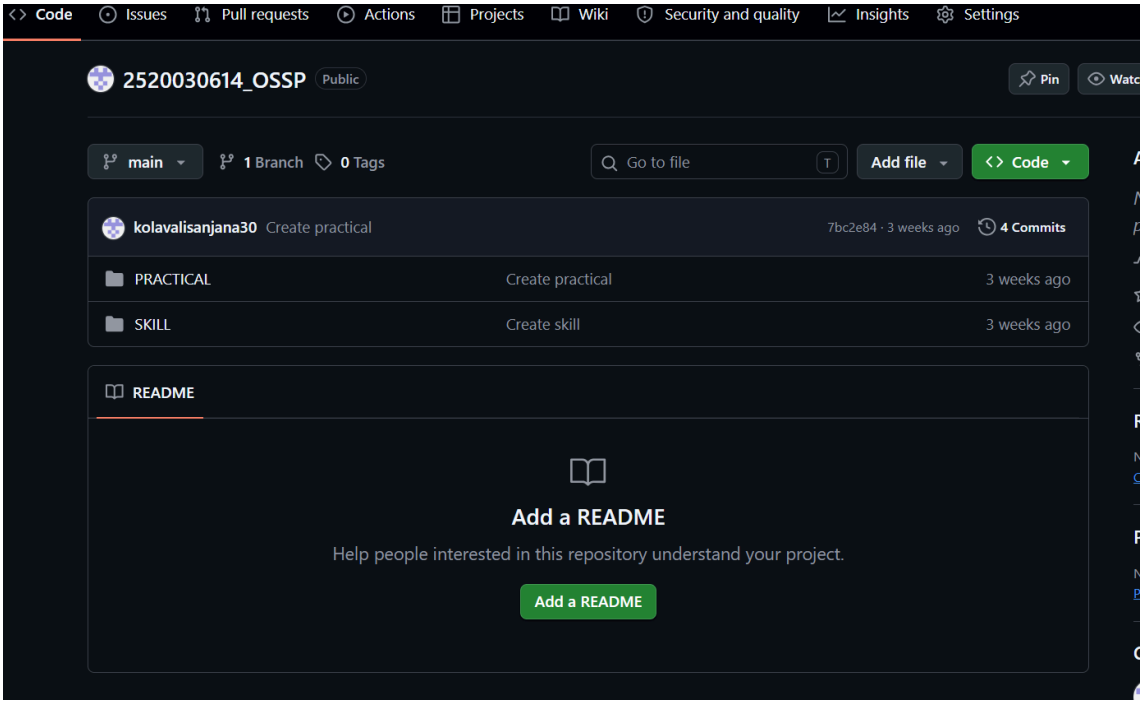
Step 9: Create Your C Program

```
sanjana3027@DESKTOP-QJF5AGU:~$ nano processid.c
```

Step 10: Compile the Program

```
sanjana3027@DESKTOP-QJF5AGU:~$ gcc processid.c -o processid
sanjana3027@DESKTOP-QJF5AGU:~$ ./processid
My Process ID (PID) =  717
sanjana3027@DESKTOP-QJF5AGU:~$
```

GITHUB STRUCTURE:



EXECUTION OF FORKPROGRAMS:

Command:

```
sanjana3027@DESKTOP-QJF5AGU:~$ nano Fork.c
```

Execution:

```
sanjana3027@DESKTOP-QJF5AGU:~$ gcc Fork.c -o Fork
sanjana3027@DESKTOP-QJF5AGU:~$ ./Fork
Hello
Hello
```

Command:

```
sanjana3027@DESKTOP-QJF5AGU:~$ nano Fork1.c
```

Execution:

```
sanjana3027@DESKTOP-QJF5AGU:~$ gcc Fork1.c -o Fork1
sanjana3027@DESKTOP-QJF5AGU:~$ ./Fork1
Parent process
Child process
```

Command:

```
sanjana3027@DESKTOP-QJF5AGU:~$ nano Fork2.c
```

Execution:

```
sanjana3027@DESKTOP-QJF5AGU:~$ gcc Fork2.c -o Fork2
sanjana3027@DESKTOP-QJF5AGU:~$ ./Fork2
x = 10
x = 10
```

Command:

```
sanjana3027@DESKTOP-QJF5AGU:~$ nano Fork3.c
```

Execution:

```
sanjana3027@DESKTOP-QJF5AGU:~$ gcc Fork3.c -o Fork3
sanjana3027@DESKTOP-QJF5AGU:~$ ./Fork3
Hello
World
C
Hello
World
C
```